Lounes Benali

Deep Learning Engineer | Speech & Multimodal AI | Large-Scale Training & Inference

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Professional Experience

AI Research Intern — Feb 2026–Aug 2026 (expected)
Institute of Foundation Models (IFM), MBZUAI — *Paris, France*

- Core contributor in a four-person team co-developing multilingual ASR; the first checkpoint ranked #1 in CER and #3 in WER on the public Open Universal ASR Leaderboard, within 2 WER points of Cohere and Audar in Arabic while supporting diacritized and code-switched transcription.
- Built a large-scale YouTube audio collection system processing 17k links/hour and nearly 4M hours total, with fault-tolerant SQLite scheduling and RAM-to-remote Opus uploads; curated 220M segments (roughly 1M hours) with multi-model signals into a high-quality 17k-hour ASR corpus.
- Ran iterative VietASR training from scratch using 100k hours of unlabeled audio for self-supervised pretraining, scaling training across 8 nodes; built Slurm workflows for distributed training and inference on AMD/ROCm and NVIDIA/CUDA clusters.
- Reimplemented the FunASR FSMN-VAD frontend and causal encoder as batched PyTorch/ROCm operations, reaching 9,668× real-time end-to-end throughput on one 8× MI210 node after initialization.

Deep Learning Engineer — Sep 2024–Feb 2026
Ampere Software Technology (Renault Group) — Master's Work-Study Program — *Guyancourt, France*

- Developed and fine-tuned TTS and speech-to-speech systems (Moshi, Orpheus, F5-TTS, ...), spanning synthetic-conversation generation, dataset processing, validation, and deployment.
- Built a real-time multi-model inference platform and Kotlin/Android Automotive clients with full-duplex capabilities, covering streaming, compression, and latency instrumentation.

Electrical Engineering Intern — Jun 2023–Aug 2023
Schneider Electric — *Algiers, Algeria*

- Studied industrial variable-frequency-drive architectures.

Education

Master's Degree in Intelligent Systems Engineering — Sep 2024–Present
Sorbonne University — *Paris, France*

Bachelor's Degree in Electrical Engineering and Computer Science — Sep 2023–May 2024
Paris-Saclay University — *Paris, France*

- Graduated top of the class.

Engineering Cycle — Electronics Engineering and Computer Science — Sep 2022–Jun 2023
École Nationale Polytechnique (ENP) — *Algiers, Algeria*

Preparatory Classes in Science and Technology — Sep 2020–Jun 2022
Two-year intensive national competitive-exam program — *Algiers, Algeria*

- Ranked 29th out of 1,576 candidates in the national competitive entrance examination for engineering schools.

Selected Technical Work

vLLM Audio-Model Integrations — IFM-Paris — GitHub

- Implemented an out-of-tree vLLM plugin for Higgs Audio v3 on ROCm/MI210, integrating its chunked Whisper frontend and stride-2 projector with native Qwen3 decoding; exposed OpenAI-compatible transcription, made configuration EngineCore-compatible, validated token/text parity against Transformers, and deployed via Slurm.

Seenitt — Streaming Visual AI for Smart Glasses — MentraOS Grant — GitHub

- Built a grant-funded smart-glasses chess assistant streaming H.264 video from Android to Python; used SAM 3/TensorRT to segment the physical board and pieces, reconstruct and track its 9×9 intersection lattice, stabilize legal states, emit FEN, and speak Stockfish guidance via Android TTS.

k2 ROCm Port for Speech Training — IFM-Paris — GitHub

- Ported upstream k2's CUDA-oriented C++/PyTorch extension to ROCm/HIP for MI210/gfx90a, adapting its build and PyTorch device, stream, allocator, and variable-length-data paths while preserving the Python API.

Technical Skills & Languages

Programming & ML: Python, PyTorch, C/C++; ASR, TTS, speech-to-speech, speech SSL, multimodal models
Distributed & Inference: Slurm, DDP/FSDP, vLLM, AMD/ROCm, NVIDIA/CUDA, TensorRT, Docker
Data & Systems: SQL, Parquet, WebDataset, WebSockets, Git
Technical Familiarity: audio encoders, codecs, continuous and discrete audio representations, CTC/RNN-T/AED, MoE, FlashAttention, flow matching, LoRA, RLHF/DPO, KV caching, PagedAttention, speculative decoding, continuous batching, quantization
Languages: French (DALF C2); English and Arabic (fluent); Berber (native)